

Reciprocating Pump Dynamics

Simulation of a Pump Mechanism Using Simscape Multibody

Machine Dynamics — Sharif University of Technology

Instructor: Dr. Sayadi

Student: Mohammadreza Sotoudeh — Student ID: 402107271

Spring 2026

1. Project Objective

The purpose of this project is to simulate a reciprocating pump mechanism using Simscape Multibody. The motor speed is set to 100 rpm and 300 rpm, and each simulation is run for 10 seconds. The required outputs are the motor torque as a function of time, the piston position inside the cylinder as a function of time, and the cylinder angle as a function of time.

2. Mechanism Model with Flywheel

The schematic of the mechanism including the flywheel is shown below.

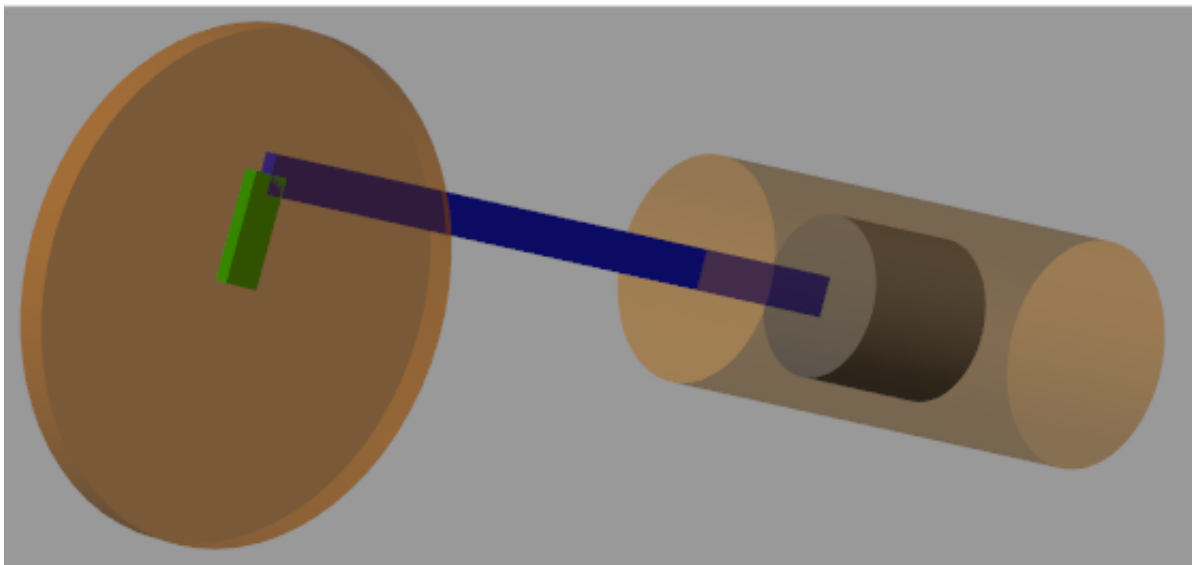


Figure 1. Simscape Multibody mechanism with flywheel.

3. Simulation at 100 rpm

The required simulation results for a motor speed of 100 rpm are presented below.

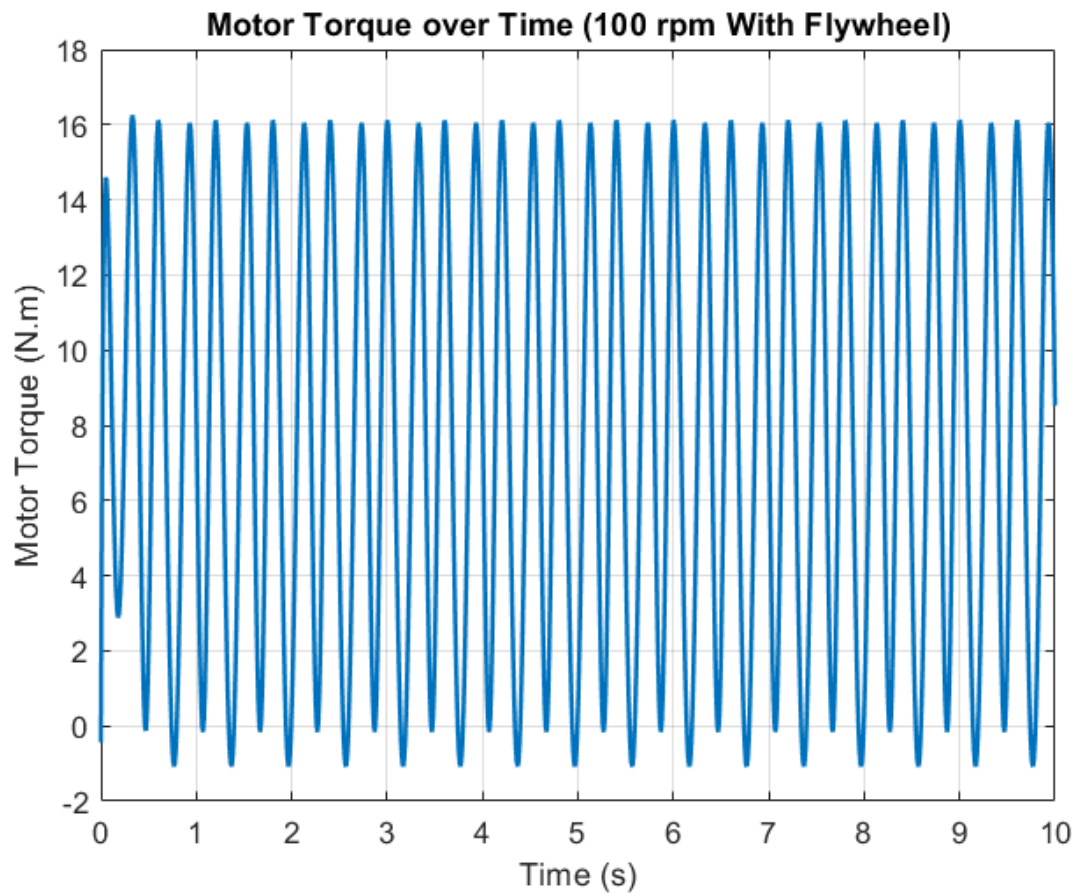


Figure 2. Motor torque versus time at 100 rpm with flywheel.

The motor torque shows a regular periodic oscillation. This behavior is expected in a crank-slider or reciprocating-pump mechanism because the load acting on the motor changes with the piston position during the forward and return strokes. The torque varies approximately from -1 N·m to 16 N·m. After an initial transient of about 0.5 s, the response becomes periodic and steady. The presence of the flywheel makes the torque fluctuations somewhat smoother.

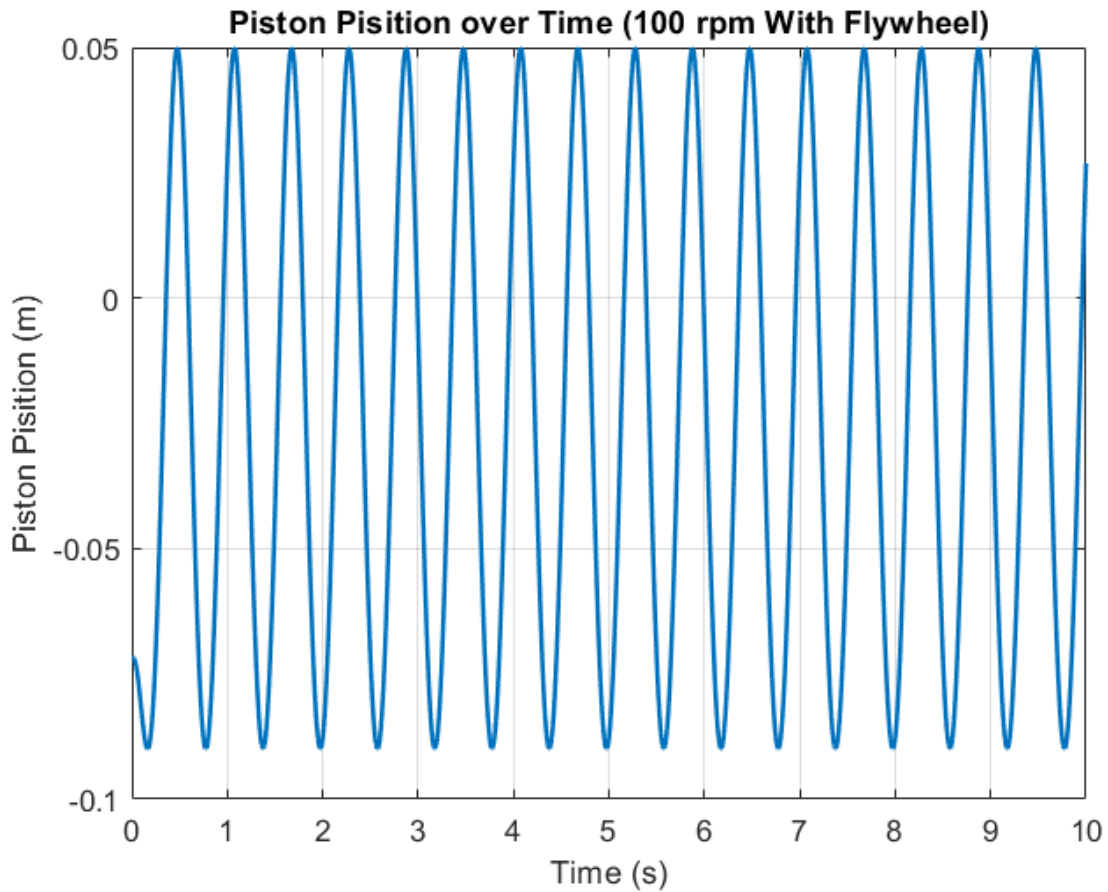


Figure 3. Piston position versus time at 100 rpm with flywheel.

At 100 rpm, the piston completes approximately 1.66 cycles per second. Therefore, one complete piston cycle takes about 0.6 s. The maximum piston position is approximately +0.05 m and the minimum is approximately -0.09 m, giving a stroke of about 0.14 m. The waveform is not a perfect sinusoid because the connecting rod causes the piston acceleration to vary near the dead-center positions.

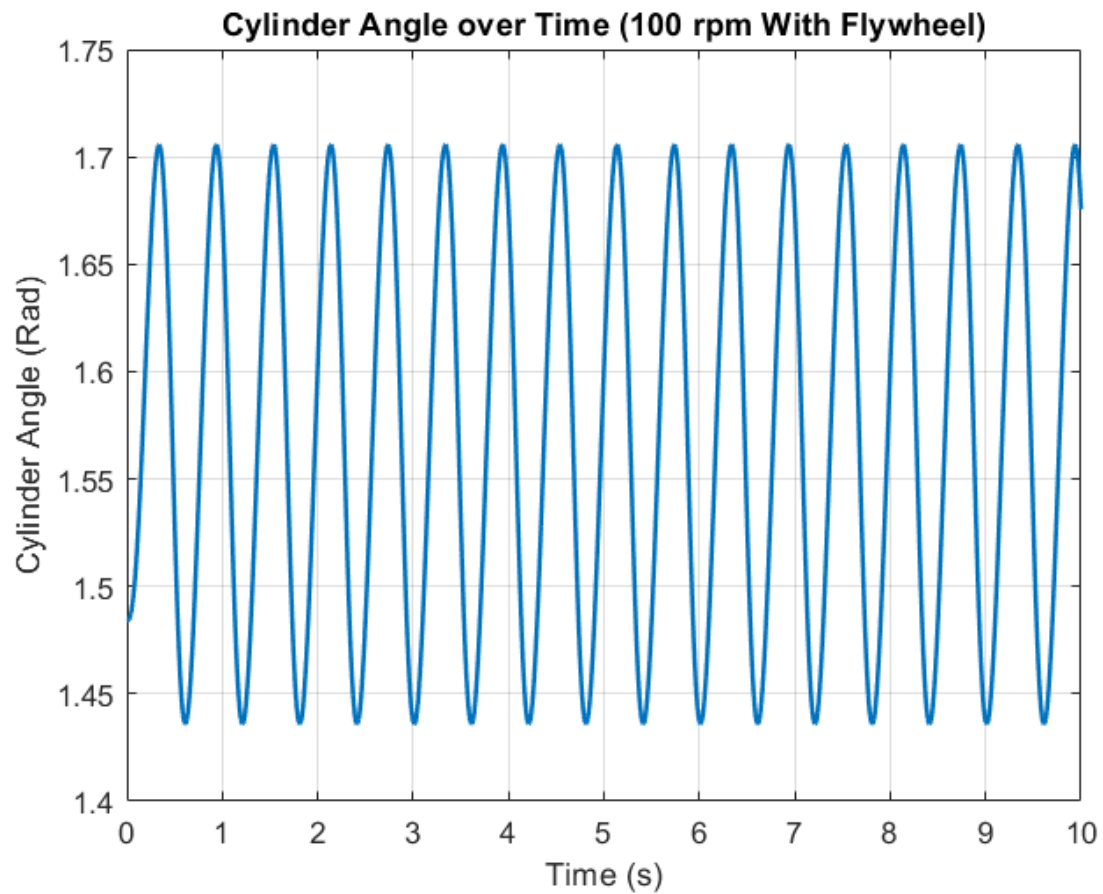


Figure 4. Cylinder angle versus time at 100 rpm with flywheel.

The cylinder angle varies periodically between approximately 1.45 and 1.71 rad. The response is regular and repeatable throughout the 10-second simulation.

4. Simulation at 300 rpm

The same mechanism is simulated at a motor speed of 300 rpm. The results are compared with the 100 rpm case.

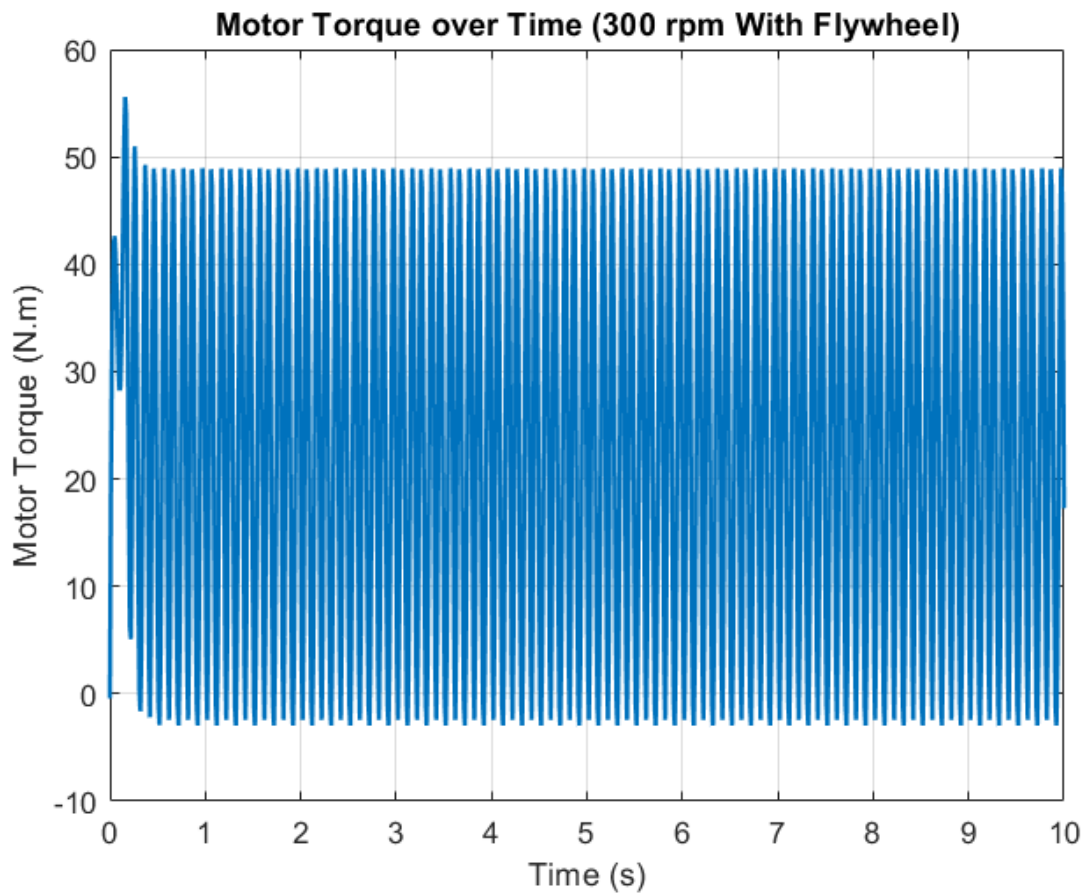


Figure 5. Motor torque versus time at 300 rpm with flywheel.

At 300 rpm, the motor torque exhibits a much higher oscillation frequency. During startup, the torque exceeds 55 N·m. After the initial transient, the torque approaches a periodic steady response, with values approximately between 0 and 50 N·m. The flywheel stores and releases energy during the cycle and helps maintain a more uniform rotational response.

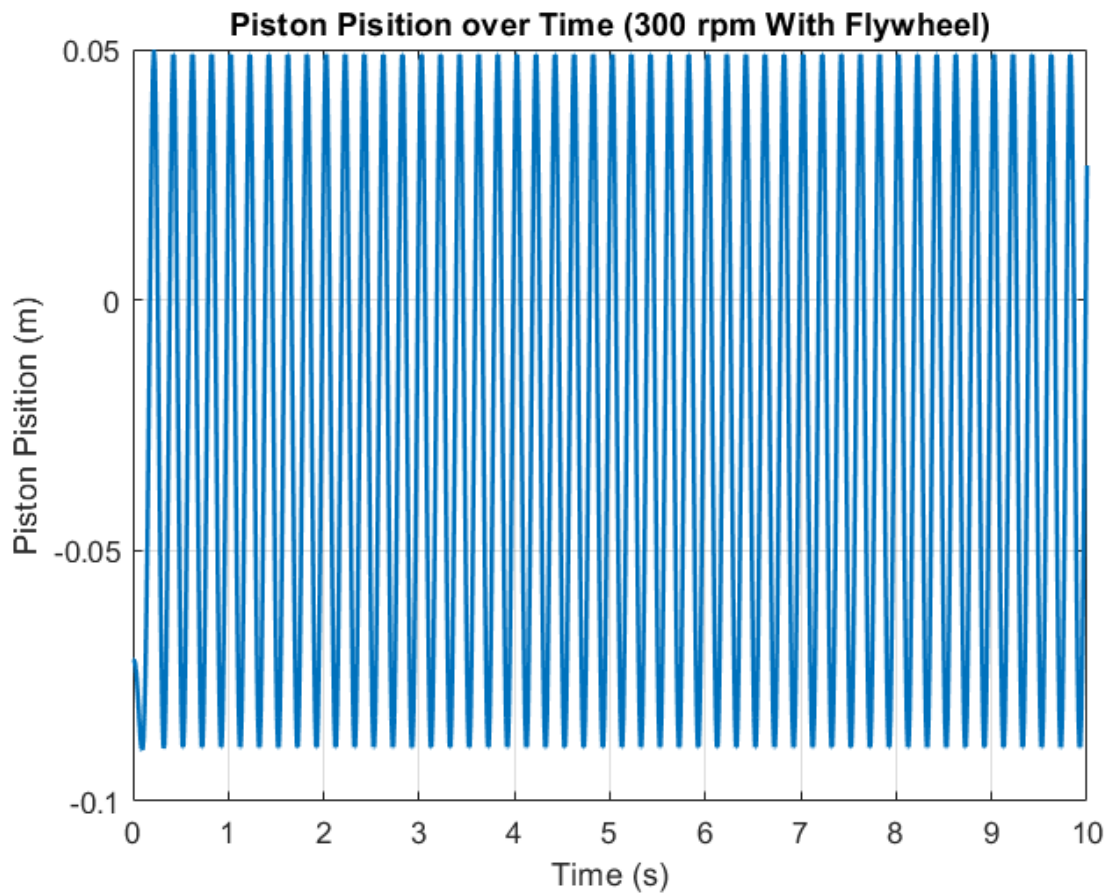


Figure 6. Piston position versus time at 300 rpm with flywheel.

The piston motion remains periodic. Its position varies approximately from -0.10 m to +0.05 m, corresponding to an approximate stroke of 0.15 m. At 300 rpm, the piston completes approximately 5 cycles per second, which is three times the frequency at 100 rpm. Although the displacement range remains approximately similar, the piston motion occurs much faster.

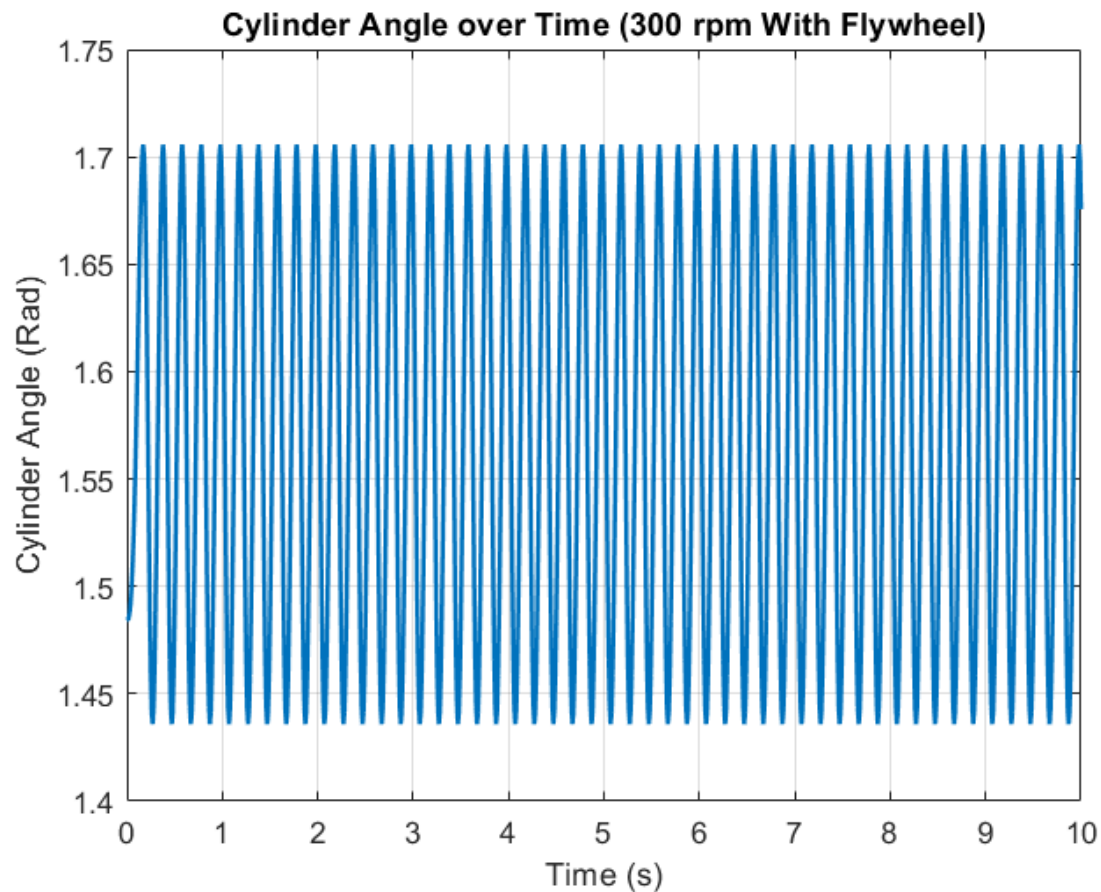


Figure 7. Cylinder angle versus time at 300 rpm with flywheel.

The cylinder angle oscillates rapidly and regularly between approximately 1.45 and 1.70 rad. The response contains approximately five complete oscillation cycles per second, consistent with the 300 rpm operating condition.

5. Comparison of 100 rpm and 300 rpm

Quantity	100 rpm	300 rpm
Simulation duration	10 s	10 s
Piston-motion frequency	≈ 1.66 cycles/s	≈ 5 cycles/s
Piston position range	≈ -0.09 to $+0.05$ m	≈ -0.10 to $+0.05$ m
Approximate stroke	0.14 m	0.15 m
Cylinder-angle range	≈ 1.45 – 1.71 rad	≈ 1.45 – 1.70 rad

Increasing the motor speed from 100 rpm to 300 rpm increases the frequency of the reciprocating motion by a factor of three. The piston displacement range remains approximately similar, while the motion and torque oscillations occur more rapidly.

6. Flywheel Effect on Motor Torque

To investigate the effect of the flywheel, the mechanism is simulated once without the flywheel and once with the flywheel. The corresponding mechanism without the flywheel is shown below.

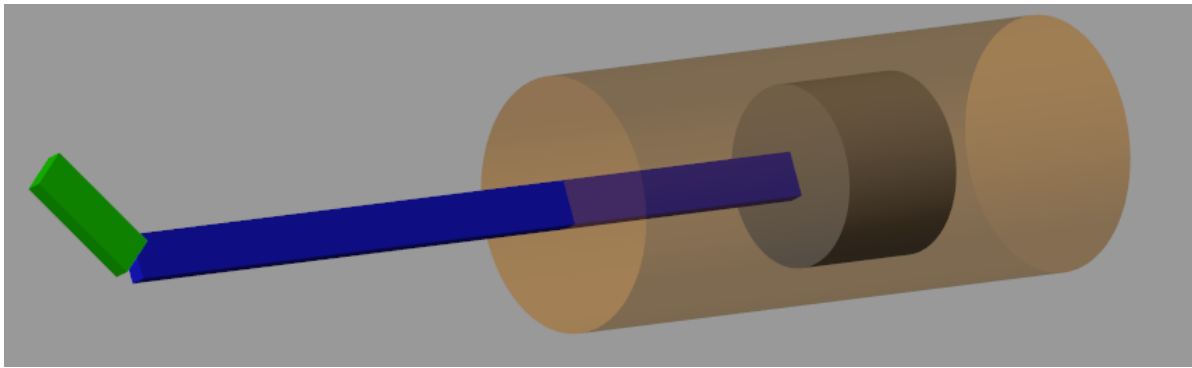


Figure 8. Mechanism configuration without flywheel.

The torque response without the flywheel is shown below.

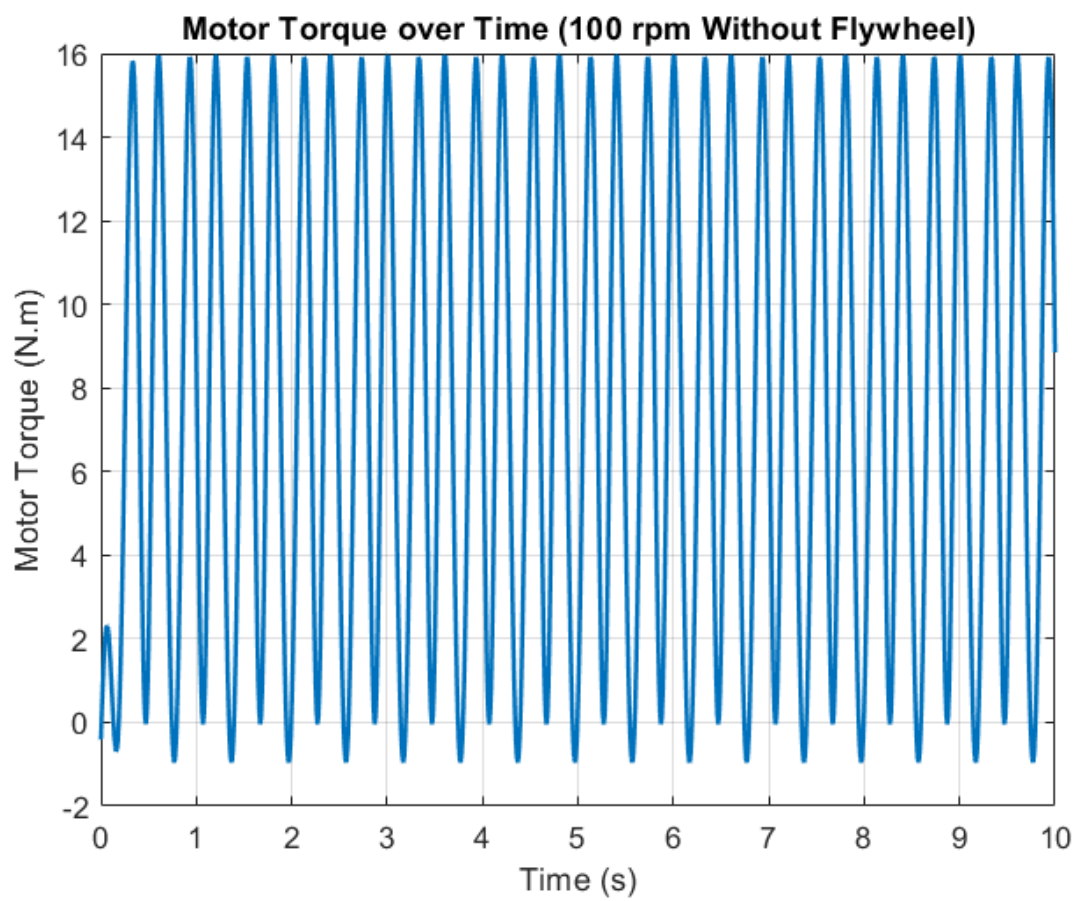


Figure 9. Motor torque versus time at 100 rpm without flywheel.

For comparison, the torque response with the flywheel is shown again below.

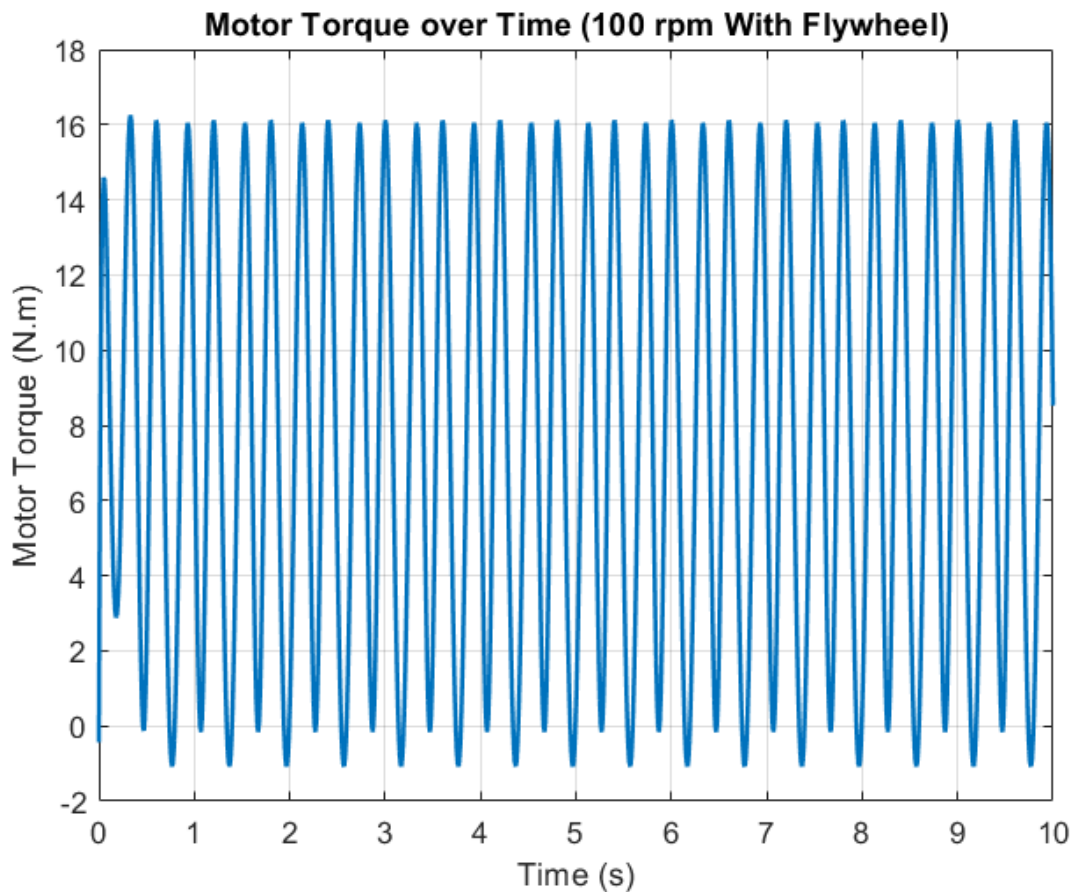


Figure 10. Motor torque versus time at 100 rpm with flywheel.

Configuration	Maximum Motor Torque
Without flywheel	16.257 N·m
With flywheel	15.960 N·m

The maximum motor torque decreases from 16.257 N·m without the flywheel to 15.96 N·m with the flywheel. This indicates that the flywheel helps absorb excess energy during lower-load portions of the cycle and return stored energy when the mechanism requires additional torque.

7. Additional Mechanism Views

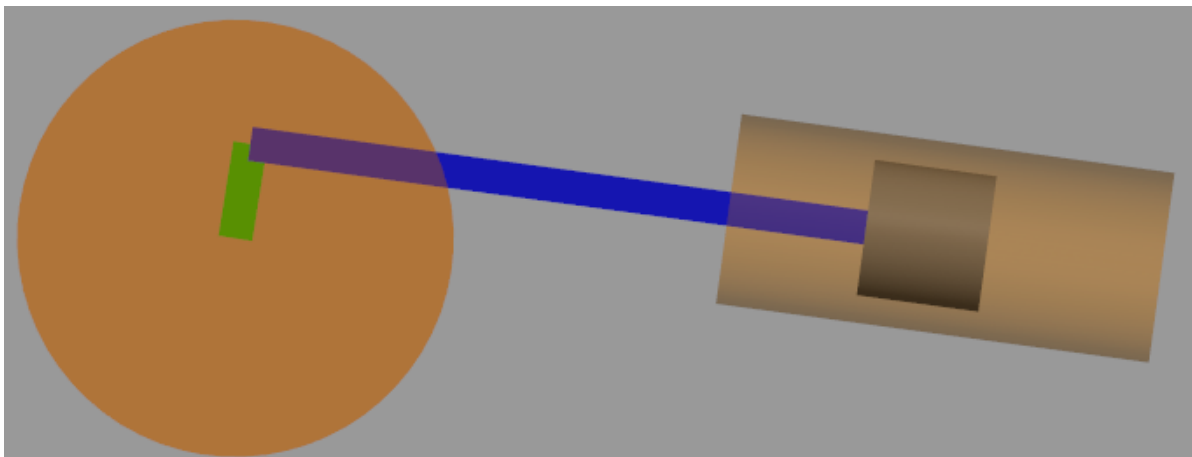


Figure 11. Additional view of the mechanism with flywheel.

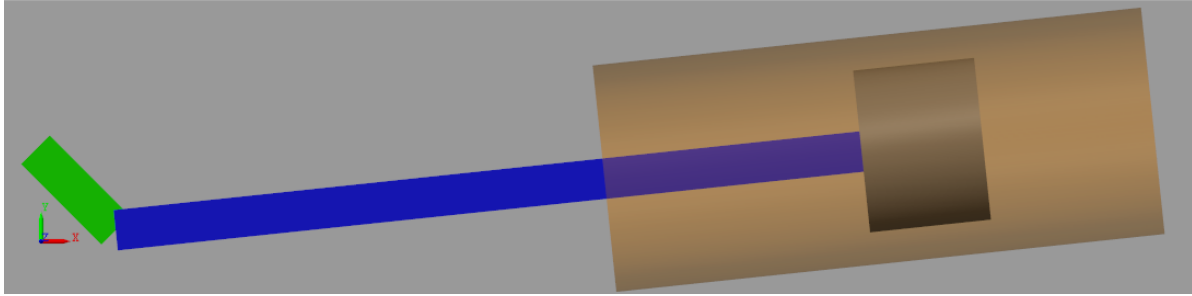


Figure 12. Additional view of the mechanism without flywheel.

8. Conclusion

The Simscape Multibody simulation demonstrates the periodic reciprocating motion of the pump mechanism at both 100 rpm and 300 rpm. The higher operating speed increases the frequency of the piston motion by a factor of three, while the displacement range remains mainly determined by the mechanism geometry. The flywheel slightly reduces the maximum motor torque from 16.257 N·m to 15.96 N·m and contributes to smoother torque behavior.